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Studierichting: Informatica

Oefeningenreeks 1C nr. 47.7

Trek de vierdemachtswortels in $\mathbb{C}$:

$$z^5 = \frac{-1 + i\sqrt{3}}{2}$$

$$\frac{-1 + i\sqrt{3}}{2} = \frac{-1}{2} + i\frac{\sqrt{3}}{2} = \text{cis}\left(\frac{2\pi}{3}\right)$$

$$\Rightarrow z_k = \text{cis}\left(\frac{\frac{2\pi}{3} + 2k\pi}{5}\right), k = 0, 1, 2, 3, 4$$

$$\Rightarrow z_k = \text{cis}\left(\frac{2\pi}{15} + \frac{2k\pi}{5}\right), k = 0, 1, 2, 3, 4$$

$$\Rightarrow z_0 = \text{cis}\left(\frac{2\pi}{15}\right)$$

$$\Rightarrow z_1 = \text{cis}\left(\frac{8\pi}{15}\right)$$

$$\Rightarrow z_2 = \text{cis}\left(\frac{14\pi}{15}\right)$$

$$\Rightarrow z_3 = \text{cis}\left(\frac{4\pi}{3}\right) = \cos\left(\frac{4\pi}{3}\right) + i\sin\left(\frac{4\pi}{3}\right) = \frac{-1}{2} - \frac{\sqrt{3}}{2}i$$

$$\Rightarrow z_4 = \text{cis}\left(\frac{26\pi}{15}\right)$$

$$\Rightarrow \boxed{z_0 = \text{cis}\left(\frac{2\pi}{15}\right), z_1 = \text{cis}\left(\frac{8\pi}{15}\right), z_2 = \text{cis}\left(\frac{14\pi}{15}\right), z_3 = \frac{-1}{2} - \frac{\sqrt{3}}{2}i, z_4 = \text{cis}\left(\frac{26\pi}{15}\right)}$$

References